The Effects of Separation and Reunion on the Behavior of Mother and Infant Squirrel Monkeys[1]

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Kaplan, Joel (1970). *The Effects of Separation and Reunion on the Behavior of Mother and Infant Squirrel Monkeys.* DEVELOPMENTAL PSYCHOBIOLOGY, 3(1): 43-52. Mother and infant squirrel monkeys that lived together in a socially restricted environment were separated for a period of seven days after the infants had become relatively independent, and were then reunited. Behavioral observations before, during, and after separation indicated that (*1*) female infants became independent of their mothers earlier than males, (*2*) neither mothers nor infants were severely affected by separation, and (*3*) an increase in attachment occurred following reunion only when the mother had a limited history of maternal experience. These results suggest that (*1*) certain characteristics in the maternal behavior of the squirrel monkey facilitate readjustment in mothers and infants following maternal separation, (*2*) maternal experience can influence the mother-infant relationship following a period of separation, and (*3*) the independence of infants from their mothers may be a function of both sex and species.

mother-infant separation reunion infant behavior maternal experience
sex and species differences squirrel monkey

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Maternal separation in macaques has been found to produce, in both mothers and infants, behaviors indicative of emotional distress (Jensen & Tolman, 1962; Seay, Hansen, & Harlow, 1962; Kaufman & Rosenblum, 1967) and resembling human responses to separation (Bowlby, 1961). However, Rosenblum and Kaufman (1968) have recently shown that different macaque species react differently to separation and reunion. They found that 6-month-old pigtail macaques (*Macaca nemestrina*) became highly disturbed when separation was imposed by removing their mothers from the social group. On the other hand, infant bonnet macaques (*M. radiata*) were relatively unaffected by the same procedures. Furthermore, when mothers and infants of both species were reunited 4 weeks later, only the pigtail macaques showed a marked increase in various measures of closeness over those observed prior to separation. Rosenblum and Kaufman (1968) have suggested that the differences found in the 2 species are related to differences in their maternal and social behaviors. For example, bonnet mothers are normally more permissive with their infants than pigtail mothers. They are less protective and punish their infants less, and regularly allow their infants to interact with other members of their group (Rosenblum & Kaufman, 1968). The importance of early social interactions was particularly stressed, since only the bonnet infants were immediately adopted by other females in their group after separation from their own mothers and tended to stay with these females even after their mothers were returned. The full significance of these social interactions in this species cannot be determined until the effects of separation and reunion have been studied under conditions that eliminate the opportunity for social contact.

The present experiment was designed to examine the effects of separation and reunion in a nonsocial environment with the squirrel monkey (*Saimiri sciureus*). In many respects, the mother-infant relationship of the squirrel monkey is very similar to that of the bonnet macaque and markedly different from that of the pigtail macaque (Rosenblum, 1968). Therefore, experimental manipulation of this relationship in the squirrel monkey might provide information that would be valuable for comparing the effects of separation and reunion.

Method

The subjects were 7 pairs of mother and infant squirrel monkeys from our breeding colony. The pairs consisted of 4 female and 3 male infants, and 5 experienced and 2 inexperienced mothers. Six of the 7 mothers had been purchased as adults from an importer; the remaining female was purchased as a juvenile. Mother and infant #35 were included in only the first part of the experiment which evaluated independence because the infant died 5 days after

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being separated from its mother. A post-mortem examination suggested that the infant died of acute pneumonia.

As pregnancies were detected, females were removed from their respective breeding pens and placed in individual cages (61 cm wide, 46 cm deep, 41 cm high), where they gave birth and remained for the duration of the experiment. Each infant remained with its mother except for a 7-day period when it was housed in a separate cage directly below and on the same rack as the mother. Therefore, visual contact between mothers and infants was prevented during the separation period, although vocal exchanges were possible.

All of the subjects were housed and observed in the same room, which was part of our general animal facility. Consequently, extraneous sounds were not entirely eliminated, but were quite minimal. Each mother and infant was observed daily for a 20-min period, beginning on the first day after birth. Using separate timers, total time spent engaging in each of the behavior categories listed below was obtained in each observation period.

Category	*Definition*
Activity	Any movement involving locomotion
Vocalization	Any vocal response that could be heard by the observer
Mother-Infant Contact	At least both legs of infant in contact with mother. Consisted mainly of infant being carried on mother's back, or in nursing position
Nipple Contact	Oral contact with the mother's nipple

The observer maintained a position approximately 13 ft from the subjects during the observation sessions. Although he made no attempt to isolate himself from the subjects, the observer made every attempt not to stare directly at the animals. Simultaneous observations of 3 pairs by visual scanning further prevented direct "full face" visual contact. Adaptation by the subjects to the presence of the observer in the room appeared to be rapid.

An infant was separated from its mother after reaching a stage where it was spending less than 10% of the time in contact with her. Specifically, a mean score of 10% or less for 5 consecutive sessions for the category of "Mother-Infant Contact" was used as the criterion for separation. (This criterion was selected since we had originally planned to focus on the reunion phase of the experiment, and any increase in mother-infant attachment after reunion would be easily detected if scores for mother-infant contact were low prior to separation.) At the time of separation each mother-infant pair was transported in

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its home cage to another room, where the infant was gently removed and placed in a separate cage. Mother and infant were then returned to the observation room in their respective cages. This procedure was used so that the other pairs of mothers and infants in the observation room would not be overly disturbed by the excitement that is normally produced when a mother and her infant are initially separated (Winter, 1968; and general observations in our own laboratory). After 7 days the infant was returned to its mother's cage in the observation room.

Results

Table 1 shows the age at which each infant was separated from its mother. With the criterion used, the data suggest that the females became independent of their mothers earlier than the males ($p<.057$, Mann-Whitney U test).

TABLE 1. Ages at which male and female infants were separated from their mother.

Infant	Sex	Separation Age (wk)	Mother's History
19	F	13	M
25	F	13	P
35	F	15	M
7	M	19	S
29	F	20	M
9	M	21	M
37	M	22	M

P = Primiparous
M = Multiparous
S = Multiparous; previous delivery stillbirth

The behavior of both mothers and infants immediately after separation was characterized by extreme activity and intense vocal shrieking, which was still evident 1 hr later (Figs. 1 and 2). Since infants were already very active before separation (significantly more than mothers, $p<.01$, analysis of variance), quantitative changes 1 hr after separation could not be detected with the measure used. However, qualitative changes were clearly observable. Only a minimal amount of vocalization accompanied activity before separation and was

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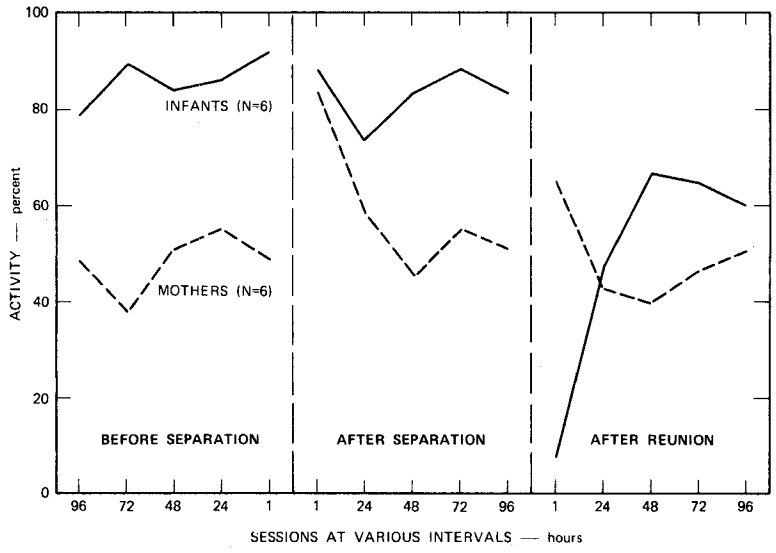


Fig. 1. Mean percent of time engaged in locomotor activity by mothers and infants at various intervals before and after separation, and after reunion.

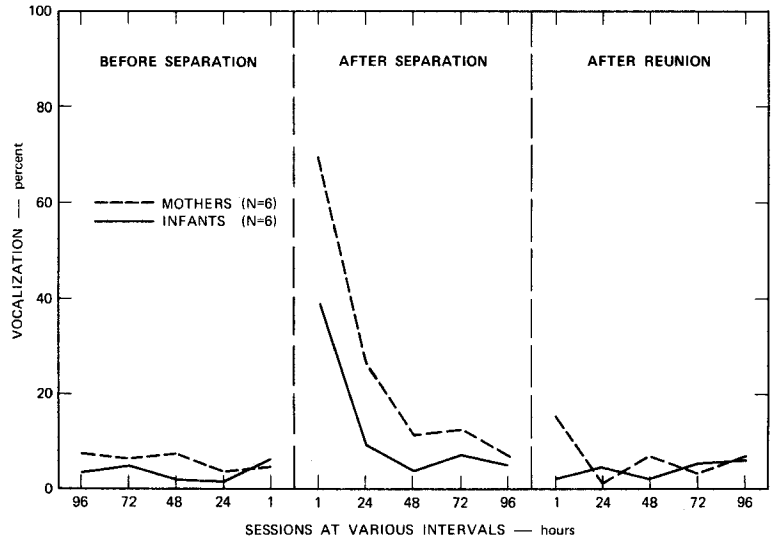


Fig. 2. Mean percent of time (total duration) of vocalization by mothers and infants at various intervals before and after separation, and after reunion.

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never the shrieking type observed after separation. Infant activity decreased somewhat 24 hr after separation and then increased again over the next 3 test sessions (Fig. 1). However, the decrease at 24 hr was not statistically significant when compared with activity on any of the other separation sessions; none of the sessions was different from those prior to separation.

Mothers showed an increase in activity ($p<.01$) and vocalization ($p<.01$) 1 hr after separation, but only in vocalization after 24 hr ($p<.05$), compared with scores obtained 1 hr before separation (t tests).

Each infant immediately climbed on its mother's back when reunited and remained there for most of the next few hours (Figs. 1 and 3). The difference in contact between the session before separation and the first session after reunion was highly significant ($p<.01$, t test). There was a large increase in variability in mother-infant contact 24 hr following reunion and thereafter, compared to the variability before separation and during the first observation period after reunion ($p<.05$ for 24 hr and 48 hr after reunion, and $p<.01$ for 72 hr and 96 hr after reunion, t tests). This variability was mainly due to the 2 pairs that included inexperienced mothers (Fig. 3). Excluding the initial reunion session where all subjects responded alike, the mother that had not given birth previously (#25) remained in contact with her infant 99% of the time, whereas the mother whose experience was limited to 1 stillbirth (#7) remained with her infant 67% of the time. Similar scores for pairs that consisted of mothers with a record of previous live births were 3%, 10%, 14%, and 23% for the same sessions. The amount of nipple contact did not change significantly after reunion but, again, the 2 infants of inexperienced mothers spent more time at the nipple than the others. No differences were found between male and female infants on any of the measures before, during, or after separation.

The behavior of mothers immediately after being reunited with their infants might be best described as one of active concern. Often a mother would reach around with an arm as if to be reassured that her infant was still on her back. Furthermore, the mothers appeared to be much more attentive to their surroundings and reacted vocally to any extraneous noise or movement. During this initial period, the mothers were also quite active, as evidenced by the increase in activity 1 hr following reunion (t test, $p<.05$ when compared with the session immediately preceding reunion; Fig. 1). A slight, although nonsignificant, increase was also observed for vocalization at that time. However, by 24 hr after reunion, activity and vocalization scores had returned to preseparation levels.

Discussion

The results of the present experiment show that maternal separation in the squirrel monkey at a stage when physical contact between the mother and infant

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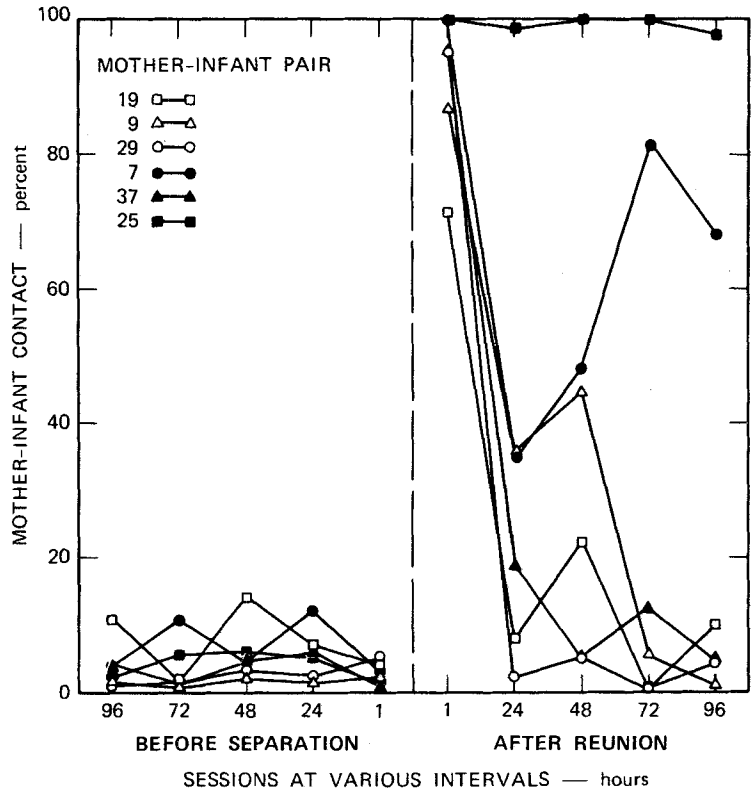


Fig. 3. Mean percent of time of physical contact between mothers and infants at various intervals before separation and after reunion.

is at a low level has only a transient effect on the behavior of mothers and infants. After an initial period of excitement following separation, the behavior of both mothers and infants returned to preseparation levels. These results are strikingly different from those found after maternal separation in certain macaques, which show signs of severe emotional depression after an initial period of excitement (Jensen & Tolman, 1962; Seay *et al.*, 1962; Kaufman & Rosenblum, 1967). However, these differences must be interpreted with caution since no attempt was made in the present experiment to separate infants at the same stage of development as used in the macaque studies. Infant macaques have usually been separated from their mothers around the age of 6 months, an age at which they are still spending a considerable amount of time in physical contact with her (Rosenblum, 1968). Our infant squirrel monkeys were separated when contact with the mother had reached a low level (less than 10% observation time). Therefore, it is possible that greater disturbances would have been produced in the present experiment had the mother and infant squirrel monkeys

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been separated at an earlier stage. This does not mean, however, that there was no longer a strong emotional attachment between our mothers and infants at the time of separation. The fact that both initially reacted strongly to separation suggests that there was still a very close tie between them. This is further supported by results showing that a squirrel monkey mother and her infant clearly recognize and prefer each other over other colony members at the same stage that separation was imposed in the present experiment (Rosenblum, 1968).

The relative lack of disturbance following maternal separation in the present experiment is similar to that reported for the bonnet macaque (Rosenblum & Kaufman, 1968). Squirrel monkey and bonnet monkey mothers are both relatively permissive with their infants (Rosenblum, 1968), which suggests that this characteristic of maternal behavior might play an important role in reducing the harmful effects of separation. In explaining the lack of disturbance observed with bonnet infants, Rosenblum and Kaufman (1968) emphasized the fact that infants were adopted by other members of their group following separation from their own mother. Our results with the squirrel monkey suggest that a lack of disturbance might not depend entirely on the effect of a substitute mother, and that bonnet infants might also be unaffected by maternal separation under conditions that prevent adoption.

Besides those aspects of maternal behavior that appear to be characteristic of certain species of nonhuman primates (e.g., permissiveness), individual differences related to a mother's history of maternal experience might also be expected to influence the outcome of separation. This was only partially the case in the present study since the inexperienced mothers and their infants behaved no differently than the other subjects during the *separation* period, but did show greater attachment than pairs containing experienced mothers during the *reunion* period.

An infant's sex did not appear to influence the results of separation or reunion in the present experiment, although differences were found in the reduction of physical contact with the mother during the normal course of development. The few studies that have been concerned with sex differences in this latter context have generally found that males become independent of their mother earlier than females (Jensen, Bobbitt, & Gordon, 1967; Mitchell, 1968). In the present experiment, the females became independent earlier than the males. It is possible that this apparent discrepancy might also be related to basic differences between species. Most of the previous reports on attachment differences in male and female infant monkeys have involved macaques, especially those known for their rigid social hierarchies and restrictive maternal attitudes. The males of these species are particularly aggressive, and since behaviors that might be considered forerunners of adult aggression are directed toward the mother early in life (Mitchell, 1968), these may promote or contribute to the earlier independence of males. Aggressive behaviors of the

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adult male are much less intense in squirrel monkeys than in macaques (Ploog, Blitz, & Ploog, 1963). In fact, outside of the mating season, adult male squirrel monkeys do not interact much with other members of their group (DuMond, 1968). It is not unlikely that these obvious differences between adult males of different species would appear early in development and affect the mother-infant relationship differentially.

The results of the present experiment with the squirrel monkey suggest that (*1*) certain characteristics of maternal behavior promote readjustment in both mothers and infants following their separation, (*2*) the mother-infant relationship after reunion is influenced by the mother's previous maternal history, and (*3*) the independence of infants from their mother may be a function of both sex and species.

Notes

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